

THREE PARADIGMS OF PHYSICAL WORLD MODELING

How should a Physical AI agent represent the environment, predict future dynamics, and survive sensory occlusion?

1. GENERATIVE PIXELS

Pixel-Space Diffusion / Autoregression

Observation Space:

Raw Pixels $\hat{l}_{t+1} \in \mathbb{R}^{H \times W \times 3}$

Forward Prediction:

Denoising U-Net / Video DiT

$$\hat{l}_{t+1} = \mathcal{D}_\theta(\epsilon, l_t, a_t)$$

Embodied Failure Modes:

- **Compute Tax:** > 30 TFLOPs/frame
- **Latency Wall:** $300\text{--}1500$ ms $\gg \tau_{\text{world}}$
- **Visual Drift:** Non-rigid morphing
- **Non-Metric:** No $SE(3)$ frame units

❌ UNVIABLE FOR REAL-TIME

2. LATENT DYNAMICS

Non-Generative JEPAs & RSSMs

Representation Space:

Latent Features $z_t \in \mathbb{R}^D$ ($D \ll HW$)

Forward Prediction:

Latent Transition $z_{t+1} = g_\phi(z_t, a_t)$
Recurrent h_t + Stochastic z_t

Embodied Advantages:

- **Sub-Millisecond:** $t_{\text{rollout}} < 2.5$ ms
- **Noise Invariance:** Discards clutter
- **Multi-Step MPC:** Rollout $H = 32$
- **Direct Planning:** Optimizes in $\mathcal{Z}$

✅ OPTIMAL FOR INTENT PLANNING

3. METRIC $SE(3)$ TREES

Rigid Lie Algebra Frame Graphs

State Space:

Homogeneous $T_B^A \in SE(3)$, $\xi \in \mathfrak{se}(3)$

Forward Prediction:

Time-Indexed Graph + EKF
 $T(t + \Delta t) = T(t) \exp(\xi^\wedge \Delta t)$

Deterministic Guarantees:

- **Microsecond Query:** $< 10 \mu\text{s}$ lookup
- **Metric Persistence:** Occlusion tracking
- **Uncertainty Bounds:** $\sigma_{\text{pos}}(t)$ growth
- **Safety Invariant:** Collision bounding

🛡️ MANDATORY FOR SAFETY

THE HYBRID STATE ARCHITECTURE OF PHYSICAL AI

The agent combines **Metric $SE(3)$ Frame Trees** (deterministic 3D spatial persistence on MCU/MPU) with **Latent World Models** (fast semantic feature rollouts on NPU), completely bypassing slow, hallucination-prone pixel diffusion.